Cybersecurity (CyberSec)

Industrial Cybersecurity.

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Agenda

1. Industrial Cybersecurity
   - Concepts
   - Critical Infrastructures
   - Critical Infrastructures
   - Problems
   - Threats, Vulnerabilities and Impact
   - Known incidents
2. Bibliography

Industrial process

- Industrial processes encompass the various operations involved in manufacturing, refining, or otherwise producing goods.
- These processes often rely on interconnected systems and technologies to function efficiently.
- Threats to these processes can include unauthorized access, tampering or sabotage. They can result in disrupted operations, compromised quality of products, or even pose risks to the workers safety.

Automation and Control Systems

- Automation refers to the use of technology to perform tasks with minimal human intervention. Control systems are integral to automation, managing and regulating processes such as temperature, pressure, or flow.
- These systems often rely on networks and communication systems for monitoring and control, making them vulnerable to cyber threats.
- Cybersecurity measures for automation and control systems aim to protect them from unauthorized access, data breaches, or malicious manipulation.

Industrial Cibersecurity

- Industrial cybersecurity involves safeguarding industrial processes, automated systems, monitoring and control systems, as well as the associated information and data, from potential threats.
- It envolves implementing security measures such as firewalls, intrusion detection systems, access controls, encryption, and regular security audits to ensure the integrity, confidentiality, and availability of industrial systems and data.

Ubiquity

- In the context of industrial cybersecurity, **Ubiquity** highlights the widespread deployment of control systems across various industrial sectors and locations.
- Ubiquitous security measures may include standardizing security protocols, implementing consistent security policies, and providing ongoing training and awareness programs to mitigate risks across diverse industrial environments.

Definition

- **Critical infrastructures** are structures or systems that support essential services necessary for the security, economy, and normal functioning of a country.
- These infrastructures are often interconnected and interdependent, meaning that disruptions or failures in one sector can have cascading effects across multiple domains, potentially leading to significant societal and economic consequences.

Examples

- Administration
- Water
- Energy
- Chemical and Nuclear Industry
- Healthcare
- Finance
- Transportation

Right and wrong

What's right

- Safety or protection against accidental events has always been considered in control systems.
- Physical security has always been considered in control systems.

What's wrong

- Cybersecurity was not traditionally considered prioritary, neither by manufacturers nor by operators.
- Regulations or standards regarding cybersecurity are not considered in control systems.

Reasons

False sense of security

- Lack of training/awareness on these topics.
- Assuming that systems operate in isolation from other systems (air gap).
- Systems used are too complex and specific to have any failure.

Reasons

Problems

- Use of off-the-shelf software.
- Systems are connected to corporate networks or the internet.
- Publicly available information about the systems used

Current situation

IT(information technologies) vs OT(operational technologies)

- Cybersecurity in industrial control systems less developed than IT cybersecurity.
- The distinctive features of control systems make it difficult to apply the same solutions directly.
- This results in: Lower level of maturity, a need for more systematic procedures, a need for adapted technologies and a need for training.

Assets

Assets are the targets that may be exposed to an attack. Some examples are:

- **Workstations** - Computers used to design, configure, and maintain system controls. They may contain sensitive information and serve as entry points for cyberattacks.
- **PLCs (Programming Logic Controllers)** - Used for controlling and automating industrial processes. They can be targets for attacks to manipulate the operation of the processes they control, which could result in material damage, production loss, or even endanger worker safety.
- **RTUs (Remote Terminal Units)** - These are systems used to control and monitor equipment remotely. If compromised, they can allow for the unauthorized manipulation of critical equipment from a distance, leading to operational disruptions or even hazardous situations.

Other Assets

Other specific assets that can be exposed to an attack:

- **HMI, SCADAS** - Human-Machine Interfaces (HMIs) and Supervisory Control and Data Acquisition (SCADA) systems are critical for monitoring and controlling industrial processes. If compromised, they can allow for the unauthorized manipulation of controlled processes, resulting in significant damage.
- **Other systems** - In addition to the mentioned examples, there are many other systems and devices that are essential for the operation of critical infrastructures. Each of these systems represents a potential target for cyberattacks, and their security must be carefully protected to ensure the integrity and operability of industrial operations.

Attack vectors

Some of the most common attack vectors are:

- **Social engineering** - Manipulating individuals into divulging confidential information or performing actions that may compromise security.
- **Spear-phishing** - Targets specific individuals or groups within an organization through personalized and convincing messages.
- **Watering hole** - Attack strategy in which an attacker guesses or observes which websites an organization often uses and infects them with malware in order to later gain control, via malware inserted on that website.
- **External staff /VPN acesss** - External staff and remote workers often require access to the organization's network or resources through virtual private network (VPN) connections. Gaining control of these may be easier.

Specific threats

- **Firmware Alteration (e.g., Stuxnet)** - Modifying the firmware so that the controlled system operates abnormally.
- **Execution Mode Alteration** - Changing the way a system operates (e.g., shutting it down).
- **Intercepting or Altering Communications** - Modifying communications with the intention of changing or disrupting normal operations.

Advanced Persistent Threats - Intro

APT are usually sophisticated cyberattacks orchestrated by individuals with considerable skill and knowledge. These attacks are typically aimed at causing operational disruptions, such as power outages or equipment failures, within targeted organizations or systems. APT attacks often involve multiple stages, including reconnaissance, initial infiltration, privilege escalation, and data exfiltration.

Advanced Persistent Threats - Definition

- **Sophisticated Attackers** - APT attackers are highly skilled and possess advanced knowledge of cybersecurity techniques and technologies. They may include state-sponsored actors, criminal organizations, or expert hacking groups.
- **Operational Disruptions** - The primary goal of APT attacks is to cause significant disruptions to the targeted systems or organizations.
- **Sophisticated Vectors** - APT attacks employ sophisticated attack vectors, including zero-day vulnerabilities, custom malware, advanced phishing techniques, and social engineering tactics.
- **Persistence and Stealth** - Once inside a target network, APT attackers often remain undetected for extended periods, stealthily exfiltrating data or carrying out malicious activities without raising suspicion.

Aurora Project - Description

The incident involved an experiment aimed at studying the feasibility of cyberattacks on control systems, particularly within the energy sector.

Aurora Project - Elaboration

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Aurora Project - Significance

This incident underscores the growing concern surrounding the cybersecurity of control systems, especially in critical sectorsm, like the energy sector. Attacks on control systems can have severe consequences, including operational disruptions, equipment damage, and even safety hazards.

Ukranian Power Grid - Description

In 2015, a cyberattack targeted the operation of the Ukrainian electrical grid, resulting in a widespread power outage lasting for 6 hours and affecting over 220,000 people.

Ukranian Power Grid - Elaboration

- **Spear-Phishing with Excel Files** - The attack began with spear-phishing emails containing Excel files embedded with malware using macro language. When employees opened these documents, their systems were infected, leading to the theft of credentials.
- **Use of Stolen Credentials** - The stolen credentials were then used to gain remote access to the system, allowing the attackers to insert commands that disrupted the normal operation of the operational infrastructure.

Ukranian Power Grid - Elaboration (cont.)

- **Attack on Gateways, Routers, and UPSs** - Additionally, gateways, routers, and Uninterruptible Power Supplies (UPSs) were targeted to prevent security teams from restoring power. The UPS system was compromised to ensure the power failure.
- **DoS Attack on Customer Service** - A Denial of Service (DoS) attack was launched on the energy company's customer service phone line. This prevented affected customers from communicating with the company to report the situation and affected areas.

Reads

- Industrial Cybersecurity [Pra23]
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Bibliography I

[Pra23] Miguel A Prada. *INDUSTRIAL CYBERSECURITY*. 2023.